

Radin Jafari

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EDUCATION

University of California, Irvine — B.S. in Electrical Engineering

Expected Jun 2027

- Coursework: Network Analysis, Digital Logic, Electromagnetics, Signals & Systems, Electronics I, II & III

Saddleback College — A.S. Physics, A.S. Mathematics

Jun 2025

- Honors: AGS Honor Society, PTK Honor Society

EXPERIENCE

Advanced Sterilization Products — Electrical Engineering Intern

Jun 2026 – Aug 2026

- Built two benchtop measurement instruments, one optical and one camera-based, from breadboard to enclosed working units.
- Wrote instrument firmware and PyQt6 operator software: serial command protocol, image processing, and SQLite logging.
- Designed an LED driver board in KiCad and brought up the assembled hardware through power-on and signal-level validation.
- Diagnosed intermittent sensor misreads as switching noise coupling onto a shared GPIO line, eliminating missed detections.
- Wound and soldered transformers and assembled bench test hardware supporting ongoing lab work.

SMART ITAC at UCI — Energy Efficiency Intern

Jan 2026 – Present

- Analyze water, gas, and electricity utility data for usage trends and savings opportunities.
- Author assessment recommendations with engineering-economic analysis of energy savings, demand reduction, and payback.
- Evaluate compressed air, lighting, and cooling measures; one lighting retrofit returned 2,818 kWh/yr at 0.7-year payback.

Zotbotics Robotics Club at UCI — Project Manager

Oct 2025 – Present

- Lead undergraduate teams building robotic arms, coordinating mechanical, electrical, and software systems.
- Authored a design requirements document and BOM for the club's arm platform, holding the build to a \$500 per-arm budget.
- Created and taught lessons on parts sourcing, BOM building, and embedded programming.

SEP Electronics Inc. — Electronics Technician

Jun 2022 – Jun 2026

- Assembled and reworked PCBs from Altium schematics, fabrication drawings, and manufacturing documentation.
- Fabricated custom wire harnesses and integrated boards, enclosures, and interconnects into finished LCD systems.

Saddleback College Robotics — Electrical Subsystem Engineer

Jan 2023 – Jun 2025

- Designed a custom CAN-transceiver PCB linking an NVIDIA Jetson Orin to the Mars rover's subsystems.
- Programmed Arduino motor diagnostics; fabricated wiring harnesses and power distribution across two competition seasons.

PROJECTS

21 cm Hydrogen Line Radio Telescope

Personal Project

- Designed a 1420 MHz receive chain: 2.2 m dish, cascaded LNAs, filtering, and image-rejection mixing to a 4 MHz IF.
- Bench-validated LNA gain, filter passband, and connector integrity at 1420 MHz with a spectrum analyzer and signal generator.
- Wrote Arduino drive firmware for the alt-az mount: joystick PWM mapping, driver fault detection, and a latched stall cutoff.

Epione — 6-Axis Robotic Arm

Team Project

- Built the electrical and software stack: Arduino Mega, dual CNC shields, TMC2209 drivers, buck-regulated servo rail.
- Wrote dual-mode control: joystick jogging plus a Python/MediaPipe gesture mode with a state machine for motor selection.
- Bridged the Python vision layer to Arduino firmware over serial, driving six steppers via AccelStepper and a servo gripper.

Cornelius — Autonomous Rover

Personal Project

- Built an ESP32-C3 rover and controller linked over ESP-NOW at 30 ms, with a 500 ms link-loss failsafe that halts the drive.
- Implemented obstacle avoidance with a servo-swept ultrasonic sensor that steers toward the side with more clearance.
- Designed the controller PCB in KiCad and the rover power path: 2S Li-ion, buck regulation, and decoupling across the motor rails.

SKILLS & AWARDS

- **Technical Skills:** Altium Designer, KiCad, LTspice, C/C++, Python, ESP32, Arduino, Raspberry Pi, RTL-SDR, Qt, OpenCV, Linux, Git
- **Awards:** Hagerty Family Osher Scholarship (\$1.5k); University Rover Challenge Finalist, top 20% of 100+ teams (2024, 2025)